

# Optimal Allocation of Observations across Run-In, Treatment, and Common Close Phases in Longitudinal Clinical Trials

Ronald G. Thomas, Ph.D.\*

June 26, 2026

## Abstract

**Background.** Trial designers planning a longitudinal study with a fixed total observation budget face an allocation question: how should observations be distributed across run-in ( $J_0$ ), treatment ( $J_1$ ), and common close ( $J_2$ ) phases to minimise the variance of the treatment-effect estimator? Existing guidance is heuristic; the analytical optimum has not been previously characterised.

**Methods.** We formulate the allocation question as a discrete optimisation problem over the Woodbury-based variance formula derived in Paper 1 of this compendium. The solution is characterised as a function of the signal-to-noise ratio  $\sigma_b/\sigma$ . Extensions cover three settings: non-equally-spaced designs in which the timing of observations within each phase is also optimised; replicated endpoint observations (closely spaced measurements at the end of treatment that average out measurement error on the final assessment, analogous to run-in averaging applied at the post-treatment end); and pattern-mixture handling of dropout following the Frost-Kenward-Fox (2008) framework.

**Results.** The optimal allocation depends primarily on  $\sigma_b/\sigma$ . The common close phase is beneficial only when this ratio exceeds a threshold that depends on  $J_1$ ; below the threshold, all observations should go to the treatment phase. Non-equally-spaced designs improve efficiency by 5-15% over equally-spaced designs at matched  $J$ , with the largest gains at low signal-to-noise ratios. Replicated endpoint observations reduce measurement-error variance proportionally to their count, with diminishing returns above 3-4 replicates.

**Conclusions.** The optimal-allocation rules derived here give trial designers a closed-form decision procedure driven by the signal-to-noise ratio. The procedure generalises to non-equally-spaced and dropout-adjusted designs without sacrificing closed-form tractability.

## Contents

### 1 Introduction

2

---

\*Wertheim School of Public Health, UCSD; ORCID: 0000-0003-1686-4965

|    |                                                                  |           |
|----|------------------------------------------------------------------|-----------|
| 37 | <b>2 Problem formulation</b>                                     | <b>3</b>  |
| 38 | 2.1 Fixed total visits . . . . .                                 | 3         |
| 39 | 2.2 Fixed total duration . . . . .                               | 4         |
| 40 | <b>3 Diminishing returns from run-in observations</b>            | <b>4</b>  |
| 41 | 3.1 Marginal value of the $J_0$ -th run-in observation . . . . . | 4         |
| 42 | 3.2 The $J_0 = 1$ anomaly . . . . .                              | 4         |
| 43 | 3.3 Correlation structure governs diminishing returns . . . . .  | 5         |
| 44 | <b>4 Optimal allocation for fixed <math>J</math></b>             | <b>6</b>  |
| 45 | <b>5 When is the common close phase beneficial?</b>              | <b>6</b>  |
| 46 | 5.1 Threshold as a function of $J_1$ . . . . .                   | 7         |
| 47 | <b>6 Non-equally-spaced designs</b>                              | <b>7</b>  |
| 48 | 6.1 General time points . . . . .                                | 7         |
| 49 | 6.2 Optimization over observation times within phases . . . . .  | 7         |
| 50 | <b>7 Replicated endpoint observations</b>                        | <b>7</b>  |
| 51 | 7.1 Motivation . . . . .                                         | 7         |
| 52 | 7.2 Variance formula with replication . . . . .                  | 8         |
| 53 | 7.3 Replication vs additional run-in observations . . . . .      | 9         |
| 54 | 7.4 Combined replication at both ends . . . . .                  | 10        |
| 55 | <b>8 Incorporating dropout</b>                                   | <b>11</b> |
| 56 | 8.1 Pattern-mixture approach . . . . .                           | 11        |
| 57 | <b>9 Numerical examples</b>                                      | <b>12</b> |
| 58 | 9.1 AD neuroimaging trial . . . . .                              | 12        |
| 59 | <b>10 Discussion</b>                                             | <b>13</b> |
| 60 | <b>11 Conflict of interest</b>                                   | <b>15</b> |
| 61 | <b>12 Funding</b>                                                | <b>15</b> |
| 62 | <b>13 Data availability statement</b>                            | <b>15</b> |
| 63 | <b>14 Reproducibility</b>                                        | <b>15</b> |
| 64 | <b>15 References</b>                                             | <b>16</b> |

## 65 1 Introduction

66 [1] Additional run-in and common close observations both  
67 improve efficiency but with diminishing or  $J_1$ -dependent re-  
68 turns.

- 69 • The first run-in observation yields the largest variance reduction.

- Subsequent run-in observations contribute progressively less.
- Common close gains depend on the number of treatment observations  $J_1$ .

To be completed by rgt.

*The relative efficiency surface in Paper 1 (Figure 1) demonstrates diminishing returns from additional run-in observations: the first run-in observation provides the largest variance reduction, while subsequent observations contribute progressively less. Similarly, common close observations improve efficiency but at a rate that depends on the number of treatment observations  $J_1$ .*

[2] **The efficiency results motivate an allocation question left open by Paper 1.**

- Trials operate under constraints on total visits or total duration.
- Paper 1 characterises efficiency but not the optimal phase split.
- The question is how to distribute observations across the three phases.

To be completed by rgt.

*These observations raise a design question that Paper 1 does not address: given operational constraints on the total number of visits or total trial duration, what is the optimal distribution of observations across the three phases?*

[3] **Prior work addresses related design questions but not optimal allocation across run-in, treatment, and common close phases.**

- Existing optima target time-point positioning or general design guidelines, not the run-in setting.
- Multiple pre-treatment measurements are known to benefit ANCOVA-analysed designs.
- Available software computes slope-comparison power but does not optimise the phase allocation.

To be completed by rgt.

<sup>1</sup> calculated optimal time-point positions for linearly divergent treatment effects under compound symmetry, but did not consider the run-in setting.<sup>2</sup> derived sample sizes for trends across repeated measurements under missing data.<sup>3</sup> showed that having more than one pre-treatment measurement is beneficial for designs analyzed via ANCOVA.<sup>4</sup> provided sample size formulas for repeated measurement designs.<sup>5</sup> developed general guidelines for the design of clinical trials with longitudinal outcomes. The `slopepower` command<sup>6</sup> and the `longpower` R package<sup>7</sup> implement power calculations for slope-comparison trials but do not optimize the allocation of observations across phases.

## 2 Problem formulation

### 2.1 Fixed total visits

[4] **The fixed-visit problem minimises the treatment-effect variance over the phase split under a total-count constraint.**

- The budget  $J$  excludes baseline and must be split as  $J_0 + J_1 + J_2 = J$ .
- Feasibility requires  $J_0 \geq 0$ ,  $J_1 \geq 1$ , and  $J_2 \geq 0$ .
- The objective is the  $\gamma\gamma$  entry of  $(X'\Sigma^{-1}X)^{-1}$ .

To be completed by rgt.

*Given  $J$  total observations (excluding baseline), choose  $(J_0^*, J_1^*, J_2^*)$  to minimize  $\text{Var}_1(\hat{\gamma})$  subject to  $J_0 + J_1 + J_2 = J$ ,  $J_0 \geq 0$ ,  $J_1 \geq 1$ ,  $J_2 \geq 0$ .*

*The objective function is  $V(J_0, J_1, J_2) = [(X'\Sigma^{-1}X)^{-1}]_{\gamma\gamma}$  computed via the Woodbury identity from Paper 1.*

## 2.2 Fixed total duration

[5] **The fixed-duration problem instead constrains total trial length and may also optimise observation timing.**

- Duration is fixed via  $T = (J_0 + J_1 + J_2) \cdot t$ , so the count constraint becomes  $J_0 + J_1 + J_2 = T/t$ .
- The variance  $\text{Var}_1(\hat{\gamma})$  is minimised subject to this constraint.
- A general version permits non-equally-spaced observations and optimises over the observation times.

To be completed by rgt.

*Alternatively, fix the trial duration  $T = (J_0 + J_1 + J_2) \cdot t$  and minimize  $\text{Var}_1(\hat{\gamma})$  subject to  $J_0 + J_1 + J_2 = T/t$ , or more generally allow non-equally-spaced observations within each phase and minimize over the observation times as well.*

## 3 Diminishing returns from run-in observations

### 3.1 Marginal value of the $J_0$ -th run-in observation

Define the marginal variance reduction from adding the  $J_0$ -th run-in observation:

$$\Delta V_{J_0} = V(J_0 - 1, J_1, J_2) - V(J_0, J_1, J_2) \quad (1)$$

### 3.2 The $J_0 = 1$ anomaly

[6] **The first run-in observation is less valuable than the second because it introduces a new model parameter.**

- For moderate  $\sigma_b/\sigma$ ,  $\Delta V_1 < \Delta V_2$ .
- Adding one run-in observation expands the model from two to three parameters, consuming a degree of freedom.

Table 1: Marginal variance reduction from the  $J_0$ -th run-in observation ( $J_1 = 2$ ,  $J_2 = 0$ ,  $\sigma^2 = 1$ ,  $\sigma_b^2 = 1$ ).

| $J_0$ | $\Delta V_{J_0}$ |
|-------|------------------|
| 1     | 0.1765           |
| 2     | 0.6667           |
| 3     | 0.5124           |
| 4     | 0.3176           |
| 5     | 0.2008           |
| 6     | 0.1342           |
| 7     | 0.0945           |
| 8     | 0.0696           |

- The second run-in observation adds no parameter and delivers pure information gain.

To be completed by rgt.

*An important non-monotonicity is visible in the table:  $\Delta V_1 < \Delta V_2$  for moderate  $\sigma_b/\sigma$ . The first run-in observation is less valuable than the second. This occurs because adding a single run-in observation forces the model from 2 parameters  $(\beta, \gamma)$  to 3 parameters  $(\delta, \beta, \gamma)$ , consuming a degree of freedom. The second run-in observation does not increase the parameter count and provides pure information gain.*

[7] **The single-run-in penalty parallels a known ANCOVA result and yields a clear design rule.**

- @frison1992 found a single pre-treatment ANCOVA covariate can reduce power at low pre-post correlation.
- The same degree-of-freedom cost applies in the run-in setting.
- Designers should plan either zero or at least two run-in observations, never exactly one.

To be completed by rgt.

*This effect is analogous to<sup>3</sup>'s finding that a single pre-treatment measurement used as an ANCOVA covariate can reduce power when the pre-post correlation is low. The practical implication is clear: a trial designer should plan either zero or at least two run-in observations, not one.*

### 3.3 Correlation structure governs diminishing returns

The correlation between the  $J_0$ -th run-in change score and the treatment-period change scores determines the information gain. Under compound symmetry:

$$\text{Corr}(C_{-J_0}, C_l) = \frac{\sigma_b^2 t^2}{2\sigma^2 + \sigma_b^2 t^2} \quad (2)$$

Table 2: Optimal allocation  $(J_0^*, J_1^*, J_2^*)$  for fixed total observations  $J$ , by signal-to-noise ratio.

| $J$ | Low ( $\sigma_b/\sigma = 0.32$ ) | Med ( $\sigma_b/\sigma = 1.0$ ) | High ( $\sigma_b/\sigma = 2.24$ ) | AD ( $\sigma_b/\sigma = 1.70$ ) |
|-----|----------------------------------|---------------------------------|-----------------------------------|---------------------------------|
| 3   | (0,3,0)                          | (0,2,1)                         | (0,2,1)                           | (0,2,1)                         |
| 4   | (0,4,0)                          | (0,2,2)                         | (0,2,2)                           | (0,2,2)                         |
| 5   | (0,5,0)                          | (0,3,2)                         | (0,3,2)                           | (0,3,2)                         |
| 6   | (0,6,0)                          | (0,3,3)                         | (0,3,3)                           | (0,3,3)                         |
| 7   | (0,5,2)                          | (0,4,3)                         | (0,4,3)                           | (0,4,3)                         |
| 8   | (0,5,3)                          | (0,4,4)                         | (0,4,4)                           | (0,4,4)                         |

[8] Under compound symmetry diminishing returns stem from the information matrix, whereas under AR(1) declining correlation accelerates them.

- The run-in change-score correlation is constant in  $J_0$  under compound symmetry.
- Diminishing returns for  $J_0 \geq 2$  therefore arise from the information matrix structure.
- Under AR(1) the correlation falls with temporal distance, yielding faster diminishing returns from distant run-in observations.

To be completed by rgt.

*which is constant in  $J_0$ . The diminishing returns (for  $J_0 \geq 2$ ) arise from the information matrix structure, not from declining correlation. Under AR(1)<sup>8</sup>, the correlation does decline with temporal distance, producing faster diminishing returns from distant run-in observations.*

## 4 Optimal allocation for fixed $J$

## 5 When is the common close phase beneficial?

[9] The common close phase aids estimation only indirectly, through the random slope, so its value is conditional.

- Common close observations inform the random slope  $b_i$ .
- They do not contribute to the treatment comparison directly.
- They are beneficial only when the added slope information sufficiently lowers  $\text{Var}(\hat{\gamma})$ .

To be completed by rgt.

*The common close phase contributes information about the random slope  $b_i$  without contributing to the treatment comparison directly. It is beneficial when the additional slope information sufficiently reduces the variance of  $\hat{\gamma}$ .*

## Minimum sigma\_b/sigma for common close benefit:

```

194 ## k=1: sigma_b/sigma >= 0.1
195 ## k=2: sigma_b/sigma >= 0.1
196 ## k=3: sigma_b/sigma >= 0.1
197 ## k=4: sigma_b/sigma >= 0.1

```

## 198 5.1 Threshold as a function of $J_1$

# 199 6 Non-equally-spaced designs

## 200 6.1 General time points

201 From the notebook derivations, the variance for the conventional design with  
202 general time points  $t_1, t_2$  (under normalization  $\sum t_i^2 = 1, \sum t_i = 0$ ) is:

$$\text{Var}(\hat{\gamma}) = \frac{3\sigma^2}{1 - t_1 t_2} + 2\sigma_b^2 \quad (3)$$

203 [10] Maximally separated time points minimise the slope-  
204 estimation variance, reproducing the classical extremes re-  
205 sult.

- 206 • The variance is minimised when  $t_1 t_2$  is most negative.
- 207 • This occurs when the two time points are maximally separated.
- 208 • It recovers the known optimality of observations at the extremes  
209 of follow-up for slope estimation.

210 To be completed by rgt.

211 *This is minimized when  $t_1 t_2$  is minimized (most negative), i.e., when the*  
212 *two time points are maximally separated. This reproduces the known result*  
213 *that observations at the extremes of the follow-up period are optimal for slope*  
214 *estimation.*

## 215 6.2 Optimization over observation times within phases

# 216 7 Replicated endpoint observations

## 217 7.1 Motivation

218 [11] The averaging logic that motivates run-in can be applied  
219 symmetrically at the end of treatment to reduce endpoint  
220 measurement error.

- 221 • Run-in averages pre-randomisation observations to cut the  
222 measurement-error contribution to the baseline slope.
- 223 • Replicated measures raise power with diminishing returns past  
224 20-50 replications.
- 225 • Closely spaced replicates of the final visit average out endpoint  
226 measurement error under the same treatment condition.

227 To be completed by rgt.

The run-in mechanism averages multiple pre-randomization observations to reduce the contribution of measurement error  $\sigma^2$  to the estimated baseline slope<sup>3,9,10</sup> showed that replicated measures increase power but with diminishing returns past 20-50 replications. The same logic applies at the end of the treatment period: if the final scheduled visit is replaced by (or augmented with) two or three closely-spaced measurements, all under the same treatment condition, the measurement error on the endpoint assessment is reduced by averaging.

**[12] Replicating an endpoint or baseline visit scales down its residual variance while leaving the random-slope contribution intact.**

- Replicating the last visit  $n_{\text{rep}}$  times over a short interval gives effective residual variance  $\sigma^2/n_{\text{rep}}$ .
- The random slope contribution  $\sigma_b^2 t^2$  is essentially unchanged.
- Replicating the baseline visit similarly reduces  $\sigma^2$  in the change-score covariance.

To be completed by rgt.

Concretely, if the last visit at time  $J_1 t$  is replicated  $n_{\text{rep}}$  times over a short interval (days to weeks, negligible relative to  $t$ ), the effective residual variance for the averaged endpoint becomes  $\sigma^2/n_{\text{rep}}$  while the random slope contribution  $\sigma_b^2 t^2$  is essentially unchanged. Similarly, the baseline visit at  $j = 0$  can be replicated to reduce  $\sigma^2$  in the change-score covariance.

**[13] Replicated endpoints differ fundamentally from common close observations in timing and the information they provide.**

- Common close observations occur at new time points after treatment stops.
- They inform the off-treatment slope.
- Replicated endpoints sit at the same time point as the last treatment visit and only reduce measurement error.

To be completed by rgt.

This is distinct from the common close design (Section 4): common close observations are at new time points after treatment stops, providing information about the off-treatment slope. Replicated endpoints are at the same time point as the last treatment visit, providing only measurement-error reduction.

## 7.2 Variance formula with replication

**[14] Replication enters only through modified entries of  $R$ , leaving the Woodbury computation otherwise unchanged.**

- The diagonal entry for a replicated visit uses  $\sigma^2/n_{\text{rep}}$  in place of  $\sigma^2$ .



Table 3: Fractional variance reduction from replicating the final endpoint ( $J_0 = 0$ ,  $J_1 = 2$ ,  $J_2 = 0$ ).

| $n_{\text{rep}}$ | $\sigma_b/\sigma = 0.5$ | $\sigma_b/\sigma = 1$ | $\sigma_b/\sigma = 2$ | $\sigma_b/\sigma = 5$ |
|------------------|-------------------------|-----------------------|-----------------------|-----------------------|
| 1                | 0.000                   | 0.000                 | 0.000                 | 0.000                 |
| 2                | 0.182                   | 0.091                 | 0.030                 | 0.005                 |
| 3                | 0.250                   | 0.125                 | 0.042                 | 0.007                 |
| 5                | 0.308                   | 0.154                 | 0.051                 | 0.009                 |

- Off-diagonal entries reflecting shared baseline error use  $\sigma^2/n_{\text{rep,pre}}$ .
- The remaining Woodbury computation proceeds identically.

To be completed by rgt.

*The residual covariance matrix  $R$  is modified so that the diagonal element corresponding to a replicated visit uses  $\sigma^2/n_{\text{rep}}$  instead of  $\sigma^2$ , and the off-diagonal elements reflecting the shared baseline error use  $\sigma^2/n_{\text{rep,pre}}$ . The Woodbury computation proceeds identically.*

[15] **Endpoint replication helps most when measurement error dominates and little when slope variability does.**

- The benefit is largest at small  $\sigma_b/\sigma$ .
- At large  $\sigma_b/\sigma > 2$  the gain is modest.
- Variance is then dominated by  $\sigma_b^2 t^2$ , which replication cannot reduce.

To be completed by rgt.

*The table confirms that endpoint replication is most beneficial when measurement error dominates ( $\sigma_b/\sigma$  small). When between-subject slope variability is large ( $\sigma_b/\sigma > 2$ ), the gain from replication is modest because the variance is dominated by  $\sigma_b^2 t^2$ , which replication cannot reduce.*

### 7.3 Replication vs additional run-in observations

[16] **A fixed budget of extra measurements can be allocated either to run-in observations or to endpoint replicates.**

- Run-in observations sit at distinct time points separated by  $t$ .
- Endpoint replicates are closely spaced at one time point.
- Replicates provide measurement-error reduction only, not slope information.

To be completed by rgt.

*A natural question is whether a fixed budget of additional measurements should be spent on run-in observations (at distinct time points, separated by  $t$ ) or on endpoint replicates (closely spaced, measurement-error reduction only).*

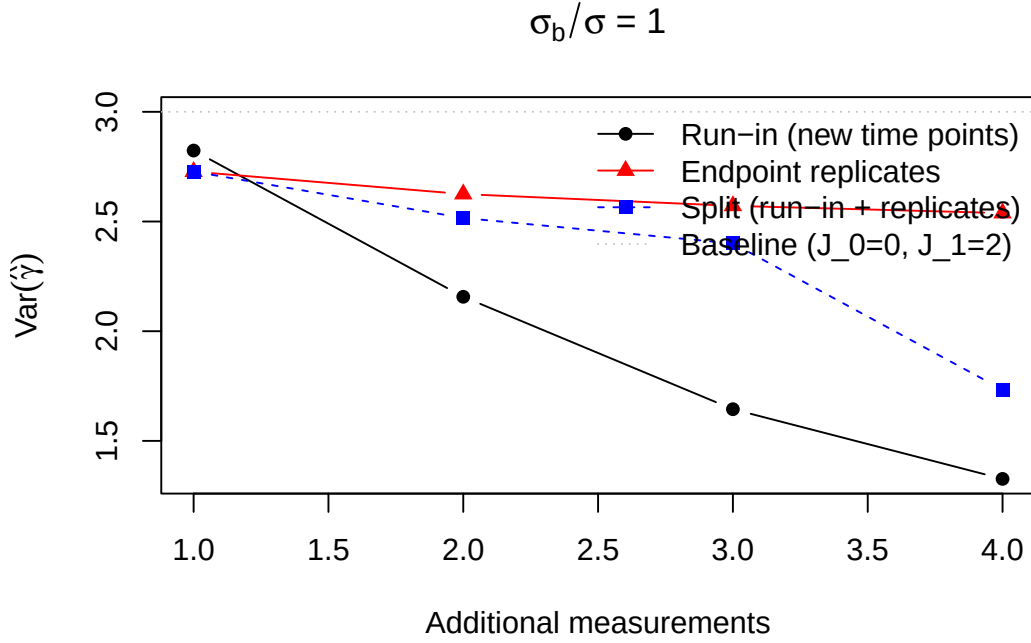


Figure 1: (`#fig:rep_vs_runin`) Variance as a function of additional measurements allocated either to run-in (new time points) or endpoint replication (same time point). Baseline:  $J_0 = 0$ ,  $J_1 = 2$ ,  $\sigma^2 = 1$ ,  $\sigma_b^2 = 1$ .

299 [17] **Whether run-in or endpoint replication wins depends on**  
300 **the signal-to-noise ratio.**

- 301 • At  $\sigma_b/\sigma \geq 1$ , run-in observations dominate.
- 302 • They supply information about the individual slope  $b_i$ , not only
- 303 measurement-error reduction.
- 304 • At  $\sigma_b/\sigma < 1$ , replication is competitive because measurement
- 305 error dominates the variance.

306 *To be completed by rgt.*

307 *When  $\sigma_b/\sigma \geq 1$ , run-in observations at new time points outperform endpoint*  
308 *replicates because they provide information about the individual slope  $b_i$ , not*  
309 *just measurement-error reduction. When  $\sigma_b/\sigma < 1$ , endpoint replication can*  
310 *be competitive because measurement error is the dominant source of variance.*

## 311 7.4 Combined replication at both ends

312 [18] **Replicating both terminal visits yields symmetric**  
313 **measurement-error reduction.**

- 314 • Baseline replication reduces  $\sigma^2$  at the start of follow-up.
- 315 • Endpoint replication reduces  $\sigma^2$  at the end of treatment.
- 316 • Together they lower measurement error symmetrically at both
- 317 ends.

318 *To be completed by rgt.*

Table 4: Per-pair variance with baseline and endpoint replication ( $J_0 = 0$ ,  $J_1 = 2$ ,  $\sigma^2 = 1$ ,  $\sigma_b^2 = 1$ ).

| $n_{\text{pre}}$ | $n_{\text{post}} = 1$ | $n_{\text{post}} = 2$ | $n_{\text{post}} = 3$ |
|------------------|-----------------------|-----------------------|-----------------------|
| 1                | 3.0000                | 2.7273                | 2.6250                |
| 2                | 2.7273                | 2.5000                | 2.4138                |
| 3                | 2.6250                | 2.4138                | 2.3333                |

Replicating both the baseline and endpoint visits provides symmetric measurement-error reduction.

## 8 Incorporating dropout

### 8.1 Pattern-mixture approach

[19] Dropout is accommodated by a pattern-mixture model that combines stratum-specific sample sizes.

- The approach follows @frost2008 Section 2.3 and related pattern-mixture work.
- Participants are stratified by their observed visit pattern.
- The overall requirement is a weighted average across strata.

To be completed by rgt.

Following<sup>9</sup> (Section 2.3), dropout is handled via a pattern-mixture model<sup>11,12</sup>. Participants are stratified by their observed visit pattern, and the overall variance is a weighted average:

$$N = \frac{1}{\sum_s p_s / N_s} \quad (4)$$

[20] The pattern-mixture formula is a conservative approximation that exact monotone-dropout treatments improve upon.

- Here  $p_s$  is the proportion with pattern  $s$  and  $N_s$  the size needed if all had pattern  $s$ .
- The approximation treats each stratum independently.
- @hu2021 derive exact formulas under monotone dropout, more rigorous and less conservative when dropout is informative about the random slope.

To be completed by rgt.

where  $p_s$  is the proportion with pattern  $s$  and  $N_s$  is the sample size that would be needed if all participants had pattern  $s$ . This approximation treats each stratum independently.<sup>13</sup> derive exact variance formulas under monotone dropout for the three-component random coefficient regression model ( $\sigma^2, \sigma_a^2, \sigma_b^2$ ), ac-

Table 5: Sample size adjusted for dropout via pattern-mixture ( $J_0 = 2$ ,  $J_1 = 2$ ,  $J_2 = 1$ , MIRIAD parameters).

| Annual dropout | $N$ (no dropout) | $N$ (with dropout) |
|----------------|------------------|--------------------|
| 0.0            | 104              | 104                |
| 0.1            | 104              | 113                |
| 0.2            | 104              | 121                |
| 0.3            | 104              | 128                |

Table 6: Top 10 allocations for  $J = 5$  observations (MIRIAD parameters,  $\Delta = 0.25$ , 90% power).

| $(J_0, J_1, J_2)$ | $\text{Var}_1(\hat{\gamma})$ | $N$ per group |
|-------------------|------------------------------|---------------|
| (0,3,2)           | 0.291783                     | 50            |
| (0,2,3)           | 0.303230                     | 51            |
| (1,2,2)           | 0.453050                     | 77            |
| (2,3,0)           | 0.551570                     | 93            |
| (0,4,1)           | 0.560024                     | 95            |
| (1,3,1)           | 0.587888                     | 99            |
| (3,2,0)           | 0.593958                     | 100           |
| (2,2,1)           | 0.616148                     | 104           |
| (0,1,4)           | 0.701145                     | 118           |
| (1,1,3)           | 0.803845                     | 136           |

counting for the effect of dropout on the estimability of the random slope variance. Their approach is more rigorous than the pattern-mixture approximation used here, which is conservative when dropout is informative about the random slope.

## 9 Numerical examples

### 9.1 AD neuroimaging trial

[21] The AD example evaluates optimal allocation under MIRIAD parameters across several budgets and dropout scenarios.

- Parameters are  $\sigma^2 = 0.3363$ ,  $\sigma_b^2 = 0.9745$ , and  $t = 1$  year.
- Allocation is compared for  $J = 4, 5, 6$  total observations.
- Both with-dropout and without-dropout cases are considered.

To be completed by rgt.

Using MIRIAD parameters ( $\sigma^2 = 0.3363$ ,  $\sigma_b^2 = 0.9745$ ,  $t = 1$  year), compare the optimal allocation for  $J = 4, 5, 6$  total observations with and without dropout.

## 363 10 Discussion

364 [22] **The optimal allocation is governed almost entirely by the**  
 365 **signal-to-noise ratio, which yields a compact design rule.**

- 366 • For most parameter configurations, one to two run-in observa-  
 367 tions and zero to one common close observation are near-optimal.
- 368 • The remaining budget should be allocated to the treatment pe-  
 369 riod, which always contributes directly to the treatment compar-  
 370 ison.
- 371 • This rule follows from diminishing returns to run-in observations  
 372 and the conditional value of the common close.

373 *To be completed by rgt.*

374 *The allocation results admit a simple summary. Across the range of signal-*  
 375 *to-noise ratios examined, including the Alzheimer’s neuroimaging regime, the*  
 376 *optimal design places one or two observations in the run-in phase, at most*  
 377 *one in the common close phase, and the remainder in the treatment period.*  
 378 *The treatment period is favored because its observations contribute directly*  
 379 *to the contrast of interest, whereas run-in and common close observations*  
 380 *contribute only indirectly, through estimation of the random slope. The near-*  
 381 *optimality of small run-in and common close counts reflects the diminishing*  
 382 *returns established in Sections 3 and 4: the first run-in observation pair*  
 383 *recovers most of the attainable variance reduction, and the marginal value of*  
 384 *additional pre- and post-treatment observations falls quickly.*

385 [23] **The signal-to-noise ratio determines whether the common**  
 386 **close phase is worth any allocation at all.**

- 387 • The common close is beneficial only when the slope-to-noise ratio  
 388 exceeds a threshold that increases with the number of treatment  
 389 observations.
- 390 • Below the threshold the entire budget, beyond a small run-in,  
 391 should go to the treatment phase.
- 392 • Designers should therefore estimate the slope-to-noise ratio from  
 393 pilot data before committing observations to a common close.

394 *To be completed by rgt.*

395 *The decision to include a common close phase is more sensitive to the vari-*  
 396 *ance parameters than the decision to include a run-in phase. Because com-*  
 397 *mon close observations inform only the random slope and never the treatment*  
 398 *contrast directly, they repay their cost only when the slope variability they help*  
 399 *estimate is itself a large component of the treatment-effect variance. This is*  
 400 *the threshold behavior of Section 4: the common close is worthwhile only*  
 401 *when  $\sigma_b/\sigma$  exceeds a bound that grows with  $J_1$ . The practical implication is*  
 402 *that the common close should be treated as an optional refinement reserved*  
 403 *for outcomes with high slope-to-noise ratios, such as neuroimaging endpoints,*  
 404 *rather than as a default design element.*

405 [24] **Two refinements, unequal spacing and endpoint repli-**

cation, offer further efficiency without changing the phase counts.

- Placing observations toward the extremes of each phase improves slope estimation, with the largest gains at low signal-to-noise ratios.
- Replicating the endpoint or baseline visit reduces measurement error and is most valuable when measurement error dominates.
- Run-in observations at new time points dominate endpoint replicates once the slope-to-noise ratio exceeds one, because they also inform the individual slope.

To be completed by rgt.

*Two refinements operate within a fixed phase allocation. First, when observation times within a phase are free, spacing them toward the extremes of the follow-up window improves slope estimation, with gains that are largest where measurement error dominates. Second, replicating the baseline or endpoint visit reduces the measurement-error contribution to the change score without adding new time points; this is most valuable at low slope-to-noise ratios and offers little when slope variability dominates. The comparison of Section 6 clarifies the choice between spending an additional measurement on a new run-in time point versus an endpoint replicate: the former dominates once  $\sigma_b/\sigma \geq 1$ , because it contributes slope information and not merely measurement-error reduction, while the latter is competitive only when measurement error is the dominant variance source.*

[25] **The allocation rules are derived under idealizing assumptions whose relaxation is the natural next step.**

- The optima assume equally spaced observations, a linear trajectory, and compound-symmetric residuals.
- Dropout is handled by a conservative pattern-mixture approximation rather than an exact monotone-missingness treatment.
- Under AR(1) residuals the diminishing returns to distant run-in observations are steeper, shifting the optimum toward fewer, closer run-in observations.

To be completed by rgt.

*Several assumptions bound the scope of these recommendations. The allocation optima are computed under equally spaced observations, linear individual trajectories, and a compound-symmetric residual structure; the AR(1) analysis of Paper 2 implies that under decaying correlation the optimum shifts toward fewer and more closely spaced run-in observations, since distant ones lose informativeness. Dropout is accommodated only through the conservative pattern-mixture approximation of Section 7, which the exact monotone-dropout treatment of<sup>13</sup> would refine. Extending the allocation optimization jointly over phase counts, within-phase timing, and the correlation structure remains the natural next step.*

## 449 11 Conflict of interest

450 [26] **The author reports no conflicts of interest.**

- 451 • No financial conflicts are present.
- 452 • No non-financial conflicts are present.
- 453 • No competing interests bear on this work.

454 To be completed by rgt.

455 *The author declares no conflicts of interest.*

## 456 12 Funding

457 [27] **The work received no specific funding.**

- 458 • No external grant supported the study.
- 459 • No internal grant supported the study.
- 460 • No sponsor influenced the analysis.

461 To be completed by rgt.

462 *This work received no specific funding.*

## 463 13 Data availability statement

464 [28] **The implementing package, tests, and shared bibliogra-**  
465 **phy are openly available in the project repository.**

- 466 • Allocation routines reside in `allocation.R`, `replicated.R`, and `de-`  
467 `sign.R`.
- 468 • Validation tests and the four-paper bibliography are included.
- 469 • Specific paths appear in the Reproducibility section below.

470 To be completed by rgt.

471 *The runinpower R package implementing the optimal-allocation routines*  
472 *(`allocation.R`, `replicated.R`, `design.R`), validation tests, and the shared*  
473 *bibliography across the four-paper compendium are openly available in the*  
474 *project repository. Specific paths are listed in the Reproducibility section*  
475 *below.*

## 476 14 Reproducibility

477 [29] **The manuscript build relies on a small, fixed set of R**  
478 **packages.**

- 479 • Rendering used R 4.4.x.
- 480 • The in-package runinpower source and tinytest provide computa-  
481 tion and testing.

482 • rmarkdown and bookdown drive the manuscript build.

483 To be completed by rgt.

484 *This manuscript was rendered with R 4.4.x. Principal packages used are the*  
485 *in-package runinpower source, tinytest (testing harness), and rmarkdown*  
486 *plus bookdown for the manuscript build.*

487 [30] **Results are deterministic, with seeding applied only**  
488 **where Monte Carlo benchmarking is used.**

- 489 • Optimisation outputs are deterministic given fixed input param-  
490 eters.
- 491 • Monte Carlo benchmarking, where reported, is reproducible.
- 492 • The driver sets `set.seed(20260418)` at its entry point.

493 To be completed by rgt.

494 *Random-number generator. Optimisation results are deterministic given*  
495 *fixed input parameters. Where Monte Carlo benchmarking is reported, the*  
496 *driver sets `set.seed(20260418)` at its entry point.*

497 [31] **Source code, companion manuscripts, and the shared bib-**  
498 **liography are located at documented repository paths.**

- 499 • The R package source is under R/.
- 500 • Companion manuscripts (Papers 1, 2, 4) are under `analysis/report/{01,02,04}-`  
501 `*/`.
- 502 • This manuscript source and the shared bibliography are under  
503 `analysis/report/03-allocation/`.

504 To be completed by rgt.

505 *Data and code availability. The R package source is at R/. The compan-*  
506 *ion manuscripts (Papers 1, 2, 4) are at `analysis/report/{01,02,04}-*`.*  
507 *The manuscript source and shared bibliography (`references.bib`, a symlink*  
508 *to `../..references.bib`) are at `analysis/report/03-allocation/`.*

## 509 15 References

- 510 1. Winkens B, Schouten HJA, Breukelen GJP van, Berger MPF. Optimal  
time-points in clinical trials with linearly divergent treatment effects.  
*Statistics in Medicine*. 2005;24(24):3743-3756. doi:10.1002/sim.2370
- 511 2. Yi Q, Panzarella T. Estimating sample size for tests on trends across re-  
peated measurements with missing data based on the interaction term  
in a mixed model. *Controlled Clinical Trials*. 2002;23(5):481-496.  
doi:10.1016/S0197-2456(02)00228-3



- 512 3. Frison L, Pocock SJ. Repeated measures in clinical trials: Analysis using mean summary statistics and its implications for design. *Statistics in Medicine*. 1992;11(13):1685-1704. doi:10.1002/sim.4780111304
- 513 4. Overall JE, Doyle SR. Estimating sample sizes for repeated measurement designs. *Controlled Clinical Trials*. 1994;15(2):100-123. doi:10.1016/0197-2456(94)90015-9
- 514 5. Galbraith S, Marschner IC. Guidelines for the design of clinical trials with longitudinal outcomes. *Controlled Clinical Trials*. 2002;23(3):257-273. doi:10.1016/S0197-2456(02)00205-2
- 515 6. Nash S, Morgan KE, Frost C, Mulick A. Power and sample-size calculations for trials that compare slopes over time: Introducing the slopepower command. *The Stata Journal*. 2021;21:572-601. doi:10.1177/1536867X211045512
- 516 7. Iddi S, Donohue MC. Power and sample size for longitudinal models in R – the longpower package and Shiny app. *The R Journal*. 2022;14(1):264-282. <https://journal.r-project.org/articles/RJ-2022-022/>
- 517 8. Winkens B, Breukelen GJP van, Schouten HJA, Berger MPF. Randomized clinical trial designs for auto-correlated longitudinal data. *Statistics in Medicine*. 2007;26(28):5023-5037. doi:10.1002/sim.2907
- 518 9. Frost C, Kenward MG, Fox NC. Optimizing the design of clinical trials where the outcome is a rate. Can estimating a baseline rate in a run-in period increase efficiency? *Statistics in Medicine*. 2008;27(18):3717-3731. doi:10.1002/sim.3280
- 519 10. Goulet MA, Cousineau D. The power of replicated measures to increase statistical power. *Advances in Methods and Practices in Psychological Science*. 2019;2:199-213.
- 520 11. Dawson JD, Lagakos SW. Size and power of two-sample tests of repeated measures data. *Biometrics*. 1993;49(4):1022-1032.
- 521 12. Lu K, Luo X, Chen PY. Sample size estimation for repeated measures analysis in randomized clinical trials with missing data. *International Journal of Biostatistics*. 2008;4(1):Article 9. doi:10.2202/1557-4679.1098
- 522 13. Hu N, Mackey H, Thomas R. Power and sample size for random coefficient regression models in randomized experiments with monotone missing data. *Biometrical Journal*. 2021;63(4):806-824. doi:10.1002/bimj.202000184

524 *Rendered on 2026-06-26 at 17:43 PDT.*

525 *Source: ~/prj/res/04-runin-power-analysis/runinpower/analysis/report/03-allocation/report.Rmd*